

The OpenFlow Protocol

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Conmutación y Enrutamiento en Redes de Datos

Reporte de Laboratorio 5

Grupo 6v

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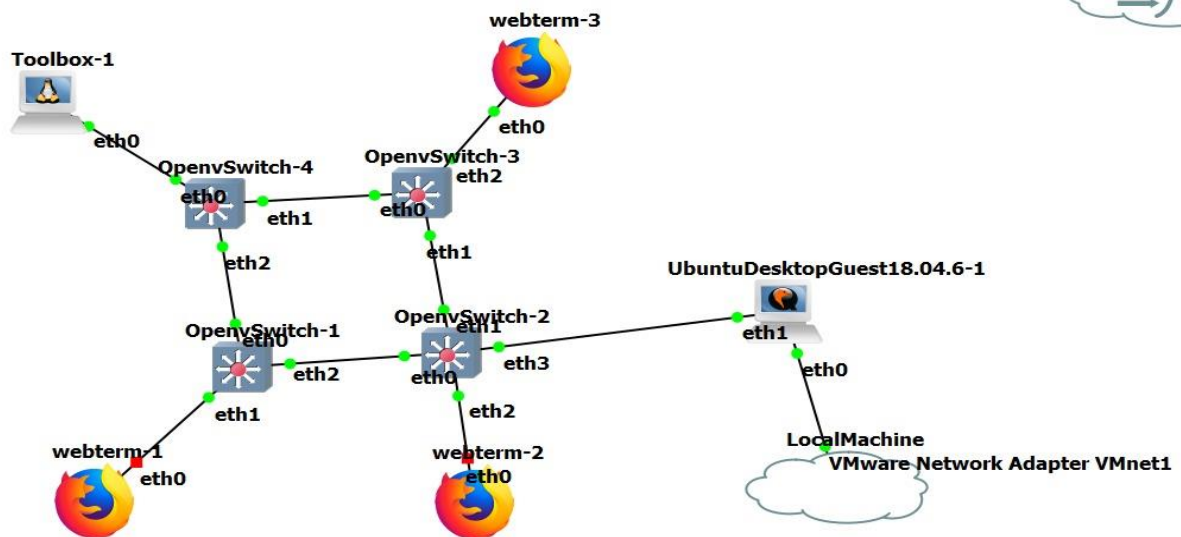
Lab 5 SDN Architecture and Flow Control with OpenVSwitch and OpenFlow

This case study explores the use of OpenVSwitch commands for virtual network management and examines the OpenFlow protocol, which enables precise control of data flows in SDN networks. It highlights how these tools and protocols facilitate flexible, centralized management of network infrastructures, while offering more granular control of network communications.

Software to Use

- GNS3
- OpenVSwitch
- Ryu Controller
- Ubuntu version 18.04
- Python y pip3
- Toolbox
- Webterm

Topology



IP Address Table

Emulated Device	Device	Interface	IP Address/Subnet Mask
Ryu Controller (Ubuntu Linux, FlowManager)	SDN-Ryu	ens3	172.16.1.10/24
Ryu Controller (Ubuntu Linux, FlowManager)	SDN-Ryu	ens4	192.168.1.10/24
Docker OpenVswitch Or Open vSwitch (v2.4.90)	OVS1	br0	192.168.1.20/24
Docker OpenVswitch Or Open vSwitch (v2.4.90)	OVS2	br0	192.168.1.21/24
Docker OpenVswitch Or Open vSwitch (v2.4.90)	OVS3	br0	192.168.1.22/24
Docker OpenVswitch Or Open vSwitch (v2.4.90)	OVS4	br0	192.168.1.23/24
Docker Webterm	PC1	eth0	192.168.1.100/24
Docker Webterm	PC2	eth0	192.168.1.110/24
Docker Webterm	PC3	eth0	192.168.1.120/24
Docker Toolbox	Toolbox-web		192.168.1.200/24
GNS3 Cloud	Local Machine	VMware Interface	172.16.1.1/24

Objectives

- Understand SDN architecture.
- Understand and use OpenVswitch commands.
- Understand the OpenFlow protocol.

The Most Important		The most complicated	
Configurar e instalar Ryu Controller en Ubuntu		Instalación de Flow manager	
Funcionamiento del OpenFlow en los OVS		Configurar Ryu Controller	
Asignación de IP a cada dispositivo			
Conexión del controlador con los dispositivos			
#	Device	Errors Description	How the identified problems were resolved
1	Ryu Controller en Ubuntu	Se intento usar la versión 24.8 de Ubuntu la cual tiene problemas de compatibilidad con Ryu Controller al utilizar una versión	En su lugar se usa la versión 18.04 con la versión de Python que viene por defecto.

		de Python que ya no es compatible.	
2	Flowmanager	No se logró la instalación de flowmanager en Ubuntu 18.04 para ver el tráfico de red de forma grafica	Flowmanager tiene conflicto al buscar las librerías y dependencias necesarias para su funcionamiento en la versión de Ubuntu que se está utilizando. Se usa línea de comandos en su lugar.
3	OpenvSwitch	No se guarda la configuración ingresada cuando se apaga/reinicia el switch. La primera solución fue editar la sección "Edit config" de cada switch en gns3, pero al hacerlo los switches dejaban de funcionar.	Se guarda, en un txt, una copia del bloque de comandos de cada switch para copiar en la consola después de encenderlos.
4	OpenVSwitch	Al ingresar el comando ovs-ofctl dump-flows br0 el switch lanzaba el error version negotiation failed (we support version 0x01, peer supports version 0x04)	Se le especifica al switch que versión debe usar con el comando ovs-ofctl -O OpenFlow13 dump-flows br0

Conclusions

Mediante el laboratorio se comprobó de manera practica el funcionamiento de SDN y su capacidad para que los switches no tomen decisiones de manera autónoma, sino que el controlador les brinde las instrucciones sobre qué hacer con cada paquete.

Gracias al uso del protocolo OpenFlow se bloquea el tráfico de protocolos específicos como por ejemplo ICMP mientras se mantienen otros como el HTTP.

El uso de comandos como `ovs-ofctl dump-flows` permitió observar el comportamiento de red y la regla que se aplica a cada paquete.

Evidences

PARTE A

Asignación de IP a los OpenvSwitch:

OvS1:

```
OpenvSwitch-1:/# ifconfig br0 192.168.1.20 netmask 255.255.255.0
OpenvSwitch-1:/# ip addr show br0
5: br0: <BROADCAST,MULTICAST,UP,LOWER_UP100> mtu 1500 qdisc
fq_codel state UNKNOWN qlen 1000
    link/ether 22:ee:d5:e1:7e:4f brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.20/24 brd 192.168.1.255 scope global br0
        valid_lft forever preferred_lft forever
    inet6 fe80::20ee:d5ff:fe1:7e4f/64 scope link
        valid_lft forever preferred_lft forever
```

OvS2:

```
OpenvSwitch-2:/# ifconfig br0 192.168.1.21 255.255.255.0

OpenvSwitch-2:/# ip addr show br0
5: br0: <BROADCAST,MULTICAST,UP,LOWER_UP100> mtu 1500 qdisc
fq_codel state UNKNOWN qlen 1000
    link/ether ca:09:2a:bb:86:40 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.21/24 brd 192.168.1.255 scope global br0
        valid_lft forever preferred_lft forever
    inet6 fe80::c809:2aff:febb:8640/64 scope link
        valid_lft forever preferred_lft forever
```

OvS3:

```
OpenvSwitch-3:/# ifconfig br0 192.168.1.22 netmask 255.255.255.0
OpenvSwitch-3:/# ip addr show br0
5: br0: <BROADCAST,MULTICAST,UP,LOWER_UP100> mtu 1500 qdisc
fq_codel state UNKNOWN qlen 1000
    link/ether 06:01:eb:a2:4e:43 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.22/24 brd 192.168.1.255 scope global br0
        valid_lft forever preferred_lft forever
    inet6 fe80::401:ebff:fea2:4e43/64 scope link
        valid_lft forever preferred_lft forever
```

OvS4:

```
OpenvSwitch-4:/# ifconfig br0 192.168.1.23 netmask 255.255.255.0
OpenvSwitch-4:/# ip addr show br0
5: br0: <BROADCAST,MULTICAST,UP,LOWER_UP100> mtu 1500 qdisc
fq_codel state UNKNOWN qlen 1000
    link/ether 1a:52:5a:93:6d:41 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.23/24 brd 192.168.1.255 scope global br0
        valid_lft forever preferred_lft forever
    inet6 fe80::1852:5aff:fe93:6d41/64 scope link
        valid_lft forever preferred_lft forever
```

Verificar que el bridge br0 exista con el comando **ovs-vsctl show**. En caso de no existir se crea utilizando el comando **ovs-vsctl add-br br0** y se asigna IP con el comando **ifconfig br0 192.168.1.20 netmask 255.255.255.0 up**.

Configuración de conexión con Ryu Controller, añadir ID único, activar Protocolo OpenFlow y Protocolo STP en cada OvS

Nota: Estos comandos se colocan en cada uno de los OvS

Comando para conectar los OvS al Ryu Controller mediante su IP 192.168.1.10:

ovs-vsctl set-controller br0 tcp:192.168.1.10:6633

Comando para añadir id único a cada OvS. En "ID" cambiar por un valor numérico único que funcionara como el ID del switch:

ovs-vsctl set bridge br0 other-config:datapath-id=0000000000000000(ID)

Comando para activar Protocolo OpenFlow en cada OvS:

ovs-vsctl set bridge br0 protocols=OpenFlow13

Comando para activar Protocolo STP en cada OvS:

ovs-vsctl set Bridge br0 stp_enable=true

INSTALACION DE RYU CONTROLLER

Actualizar Ubuntu y pip de Python

sudo apt update

sudo apt-get upgrade

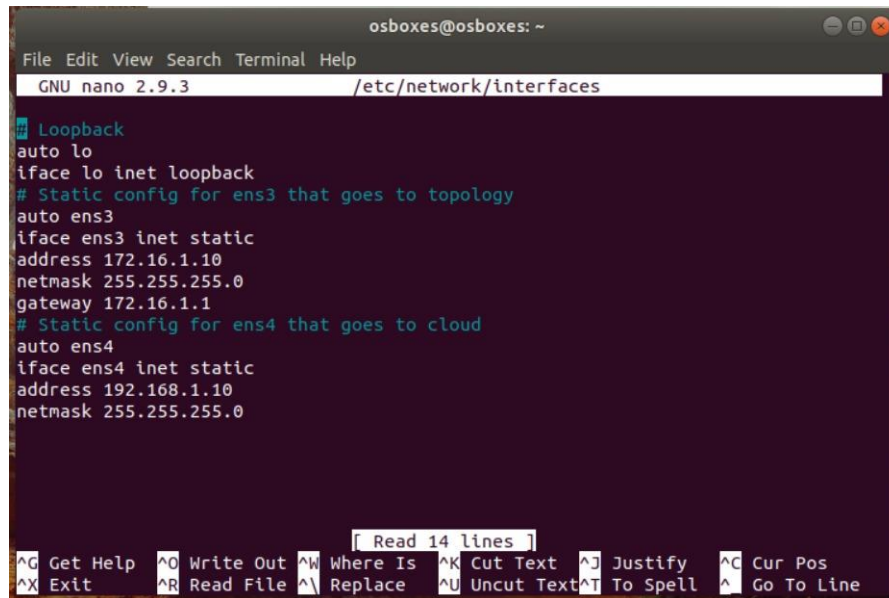
sudo apt-get install python3-pip

Instalar Ryu Controller

sudo pip3 install ryu

Verificar instalación con comando **ryu-manager --version**

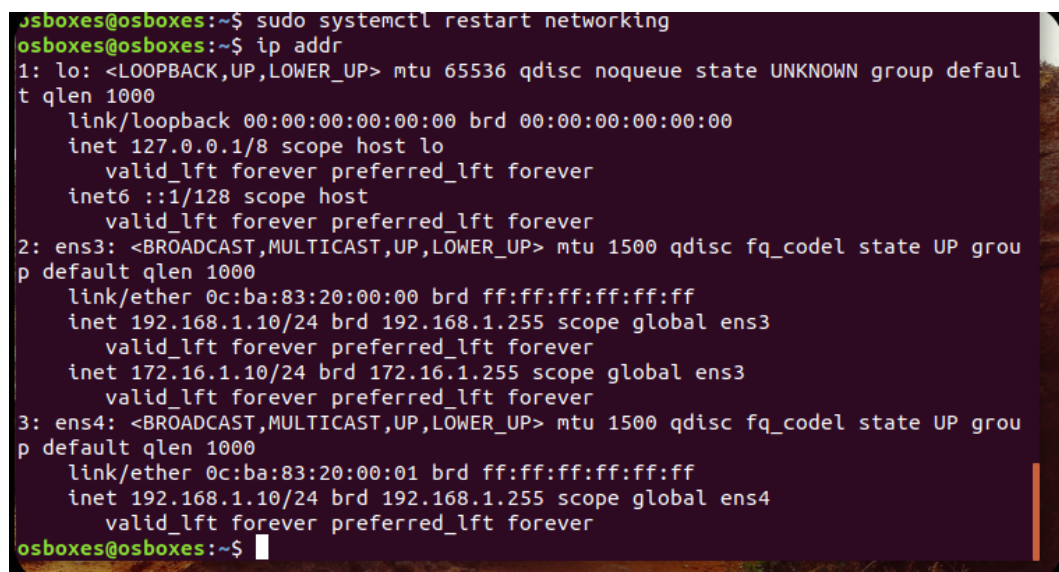
Configuración de IP para los puertos de SDN-Ryu:
Entrar al archivo de configuración de interfaces de red y editarlo con las IP correspondientes a la tabla de enrutamiento.
Comando: **nano /etc/network/interfaces**



```
osboxes@osboxes: ~  
File Edit View Search Terminal Help  
GNU nano 2.9.3 /etc/network/interfaces  
# Loopback  
auto lo  
iface lo inet loopback  
# Static config for ens3 that goes to topology  
auto ens3  
iface ens3 inet static  
address 172.16.1.10  
netmask 255.255.255.0  
gateway 172.16.1.1  
# Static config for ens4 that goes to cloud  
auto ens4  
iface ens4 inet static  
address 192.168.1.10  
netmask 255.255.255.0  
[ Read 14 lines ]  
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos  
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line
```

Despues ctrl + o, enter y ctrl + x para guardar la configuracion.

Reiniciar ajustes de red con comando **sudo systemctl restart networking**
Verificación con comando **ip addr**



```
osboxes@osboxes:~$ sudo systemctl restart networking  
osboxes@osboxes:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 0c:ba:83:20:00:00 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.10/24 brd 192.168.1.255 scope global ens3  
        valid_lft forever preferred_lft forever  
    inet 172.16.1.10/24 brd 172.16.1.255 scope global ens3  
        valid_lft forever preferred_lft forever  
3: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 0c:ba:83:20:00:01 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.10/24 brd 192.168.1.255 scope global ens4  
        valid_lft forever preferred_lft forever  
osboxes@osboxes:~$
```

Se prueba que Ubuntu tenga conexión con los switches mediante un ping a cada uno.

```
osboxes@osboxes:~$ ping 192.168.1.21
PING 192.168.1.21 (192.168.1.21) 56(84) bytes of data.
64 bytes from 192.168.1.21: icmp_seq=1 ttl=64 time=52.7 ms
64 bytes from 192.168.1.21: icmp_seq=2 ttl=64 time=21.0 ms
64 bytes from 192.168.1.21: icmp_seq=3 ttl=64 time=20.9 ms
64 bytes from 192.168.1.21: icmp_seq=4 ttl=64 time=15.8 ms
64 bytes from 192.168.1.21: icmp_seq=5 ttl=64 time=18.0 ms
64 bytes from 192.168.1.21: icmp_seq=6 ttl=64 time=18.5 ms
64 bytes from 192.168.1.21: icmp_seq=7 ttl=64 time=15.9 ms
64 bytes from 192.168.1.21: icmp_seq=8 ttl=64 time=24.0 ms
64 bytes from 192.168.1.21: icmp_seq=9 ttl=64 time=15.4 ms
64 bytes from 192.168.1.21: icmp_seq=10 ttl=64 time=18.8 ms
64 bytes from 192.168.1.21: icmp_seq=11 ttl=64 time=39.2 ms
64 bytes from 192.168.1.21: icmp_seq=12 ttl=64 time=20.8 ms
64 bytes from 192.168.1.21: icmp_seq=13 ttl=64 time=26.3 ms
^C
--- 192.168.1.21 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12041ms
rtt min/avg/max/mdev = 15.463/23.691/52.763/10.337 ms
osboxes@osboxes:~$
```

Encender el controlador con el comando
ryu-manager ryu.app.simple_switch_13

Si Ryu Controller se conecta correctamente con los switches debe mostrar la siguiente información.

```
osboxes@osboxes:~$ ryu-manager ryu.app.simple_switch_13
loading app ryu.app.simple_switch_13
loading app ryu.controller.ofp_handler
instantiating app ryu.app.simple_switch_13 of SimpleSwitch13
instantiating app ryu.controller.ofp_handler of OFPHandler
packet in 2 0c:ba:83:20:00:01 33:33:00:00:00:fb 4
packet in 2 0c:ba:83:20:00:01 01:00:5e:00:00:fb 4
packet in 3 0c:ba:83:20:00:01 33:33:00:00:00:fb 2
packet in 3 0c:ba:83:20:00:01 01:00:5e:00:00:fb 2
packet in 4 0c:ba:83:20:00:01 33:33:00:00:00:fb 2
packet in 4 0c:ba:83:20:00:01 01:00:5e:00:00:fb 2
packet in 1 0c:ba:83:20:00:01 33:33:00:00:00:fb 1
packet in 1 0c:ba:83:20:00:01 01:00:5e:00:00:fb 1
packet in 4 22:ee:d5:e1:7e:4f ff:ff:ff:ff:ff:ff 3
packet in 3 22:ee:d5:e1:7e:4f ff:ff:ff:ff:ff:ff 1
packet in 3 06:01:eb:a2:4e:43 22:ee:d5:e1:7e:4f 4294967294
packet in 2 22:ee:d5:e1:7e:4f ff:ff:ff:ff:ff:ff 2
packet in 4 22:ee:d5:e1:7e:4f ff:ff:ff:ff:ff:ff 3
```

Con el comando **ovs-ofctl -O OpenFlow13 dump-flows br0** podemos verificar que las reglas del controlador se están aplicando a los OvS:

```
OpenvSwitch-1:~# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=181.138s, table=0, n_packets=180, n_bytes=9540, priority=0
actions=CONTROLLER:65535
```

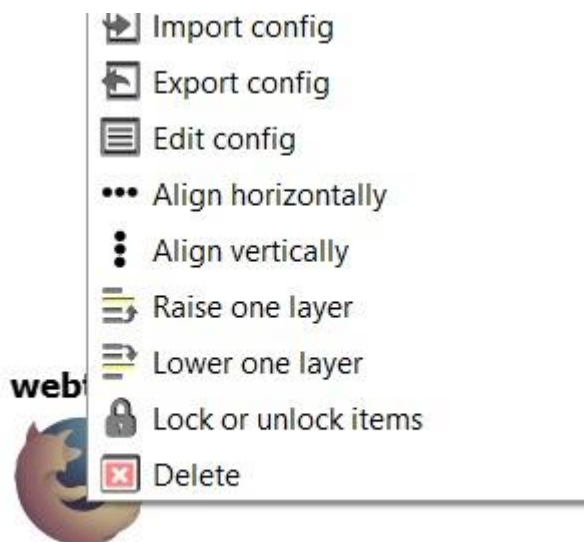
Asignación de IP a cada PC (webterm)

Se utiliza el comando **ip addr add (IP/Mask) dev (Puerto)**, se enciende la interfaz con el comando **ip link set (puerto) up** y verificamos con el comando **ip addr show (puerto)**.

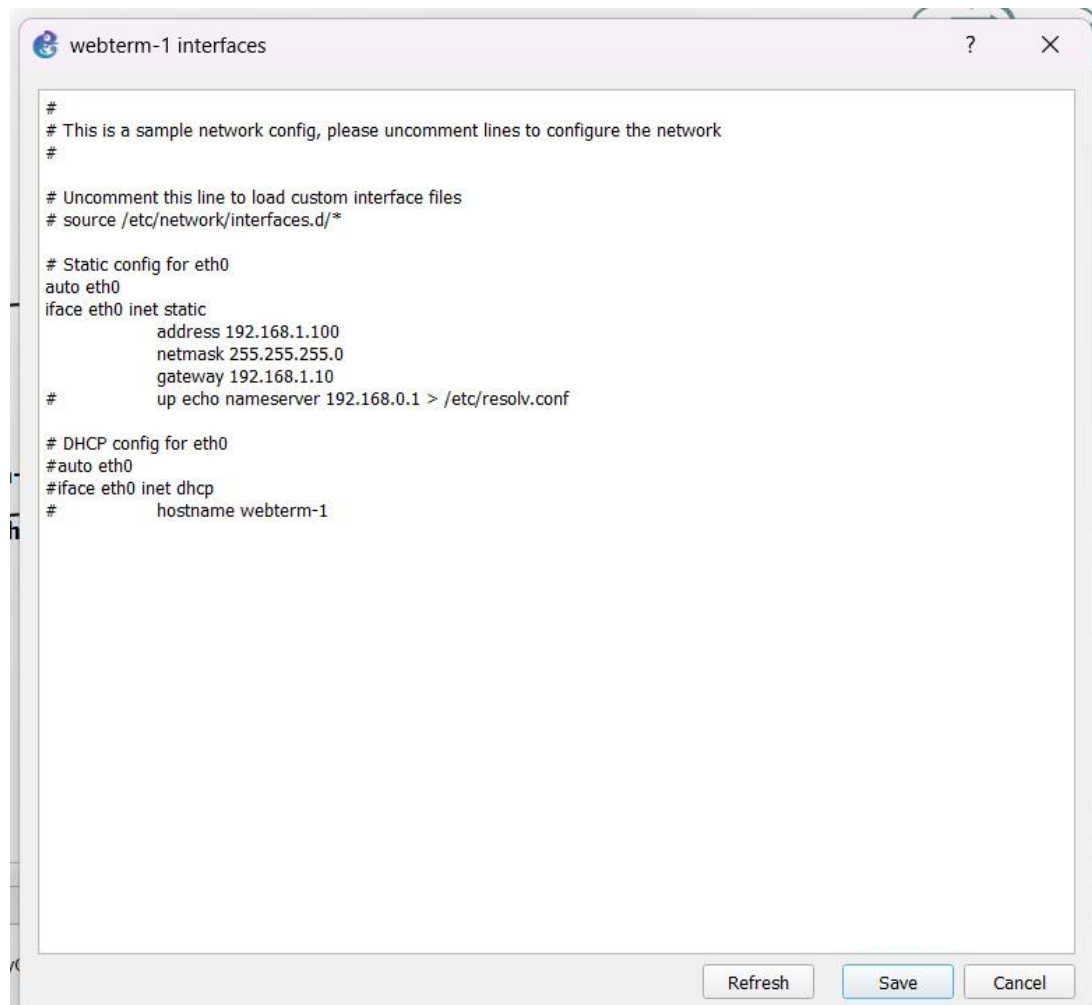
PC1

```
root@webterm-1:~# ip addr add 192.168.1.100/24 dev eth0
root@webterm-1:~# ip link set eth0 up
root@webterm-1:~# ip addr show eth0
214: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOW
WN group default qlen 1000
    link/ether 02:42:f9:76:1b:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:f9ff:fe76:1b00/64 scope link
        valid_lft forever preferred_lft forever
root@webterm-1:~#
```

Para asegurar que la configuración se guarde y aplique cada que se enciendan las PC se configura el apartado “edit config”.



Y se coloca la IP, Mask y Gateway de la siguiente manera (El símbolo # sirve para comentar la línea escrita).



Damos “save” y repetimos el proceso con las demás PC

PC2:

```
# Static config for eth0  
auto eth0  
iface eth0 inet static
```

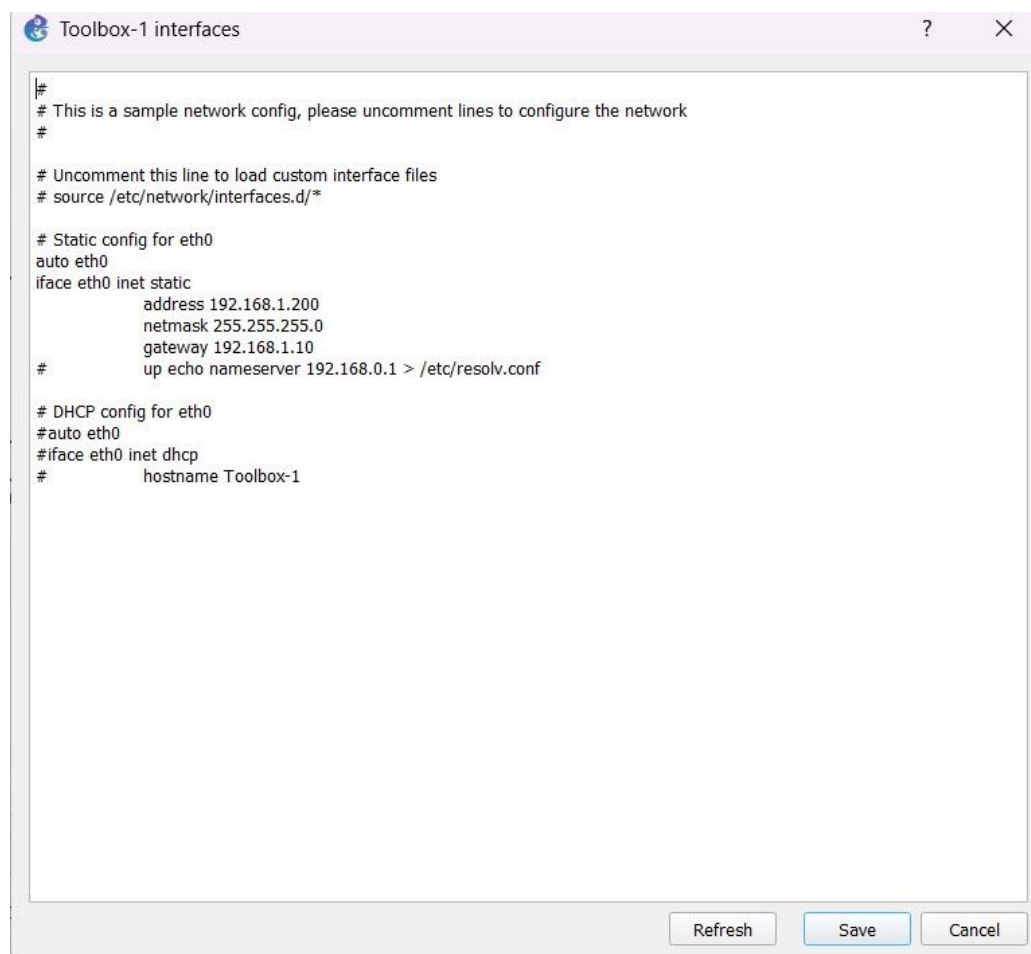
```
    address 192.168.1.120  
    netmask 255.255.255.0  
    gateway 192.168.1.10
```

PC3:

```
# Static config for eth0
auto eth0
iface eth0 inet static
```

```
address 192.168.1.110
netmask 255.255.255.0
gateway 192.168.1.10
```

Configuramos la IP de Tool-box de la misma manera que las PC para asegurarnos que la configuración se guarde en cada arranque de Tool-box



PARTE B

Recall the Architecture and Components of an OpenFlow Architecture

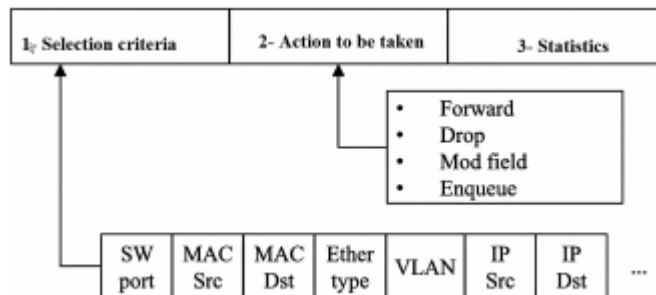
1. Recall the three essential components of an OpenFlow architecture.
 - The controller: Habilita las decisiones de enrutamiento y la administración de redes.
 - OpenFlow Switches: Estos dispositivos implementan las reglas establecidas por el controlador.
 - The flow table: Estas comprenden un conjunto de reglas e instrucciones que son implementadas por el switch.

2. Link each item in the table with its corresponding definition.

1. Flow Table	A. It allows you to manage multicast and broadcast traffic more efficiently. It can be used to redirect packets to multiple destinations or to apply specific filtering rules.
2. Group Table	B. It collects information on the traffic passing through the switch. This information can be used to monitor the network, control quality of service, or optimize bandwidth utilization.
3. Statistics Table	C. It is a list of rules that determine how the switch should handle packets. These rules can be based on criteria such as source or destination IP address, TCP or UDP port, or data type.

- Flow Table → inciso C
- Group Table → inciso A
- Statistics Table → inciso B

3. Recall three advantages of an OpenFlow architecture:
 - Permite gestionar la red desde un solo punto.
 - Permite crear reglas personalizadas.
 - Funciona en switches de distinto fabricante que soporten openflow.
4. Explain components 1, 2, and 3 of a Flux Input:



1. Compara los datos del encabezado del paquete para ver si coincide con una instrucción específica.
2. Acción que se toma con el paquete dependiendo las instrucciones dadas.
3. Colecta la información sobre el tráfico que pasa por el switch.

PARTE C

1. Display the configuration of the OVS1 switch. On the OVS1 switch, use the command: `ovs-vsctl show`

```
OpenvSwitch-1:/# ovs-vsctl show
fc366f4b-d070-4479-ac20-fd0bd751a67a
```

```
Bridge br2
  datapath_type: netdev
  Port br2
    Interface br2
      type: internal
Bridge br3
  datapath_type: netdev
  Port br3
    Interface br3
      type: internal
Bridge br1
  datapath_type: netdev
  Port br1
    Interface br1
      type: internal
Bridge br0
  Controller "tcp:192.168.1.10:6633"
    is_connected: true
  datapath_type: netdev
  Port eth2
    Interface eth2
  Port eth12
    Interface eth12
  Port eth3
    Interface eth3
  Port eth0
    Interface eth0
  Port eth11
    Interface eth11
  Port br0
    Interface br0
      type: internal
  Port eth6
    Interface eth6
  Port eth14
    Interface eth14
  Port eth1
    Interface eth1
  Port eth10
```

```
Interface eth10
Port eth9
Interface eth9
Port eth7
Interface eth7
Port eth15
Interface eth15
Port eth5
Interface eth5
Port eth4
Interface eth4
Port eth13
Interface eth13
Port eth8
Interface eth8
```

2. . Display the flow table of switch OVS1. On the OVS1 switch, use the command: `ovs-ofctl dump-flows br0`

```
OpenvSwitch-1:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=423.047s, table=0, n_packets=452,
n_bytes=24232, priority=0 actions=CONTROLLER:65535
OpenvSwitch-1:/#
```

3. . Referring to the output of this command, associate each field with its corresponding field in the following table:
cookie=0x0, duration=44.013s, table=0, n_packets=33, n_bytes=2170, priority=1, in_port=eth2, dl_src=de:bf:a3:9b:4e:1b, dl_dst=06:7a:39:56:3c:ed actions=output:eth1

- Flow Table number: table=0
- Criteria priority: priority=1
- Selection criteria: in_port=eth2, dl_src=de:bf:a3:9b:4e:1b, dl_dst=06:7a:39:56:3c:ed
- Action to be taken: actions=output:eth1
- Number of packages matching this criterion: n_packets=33

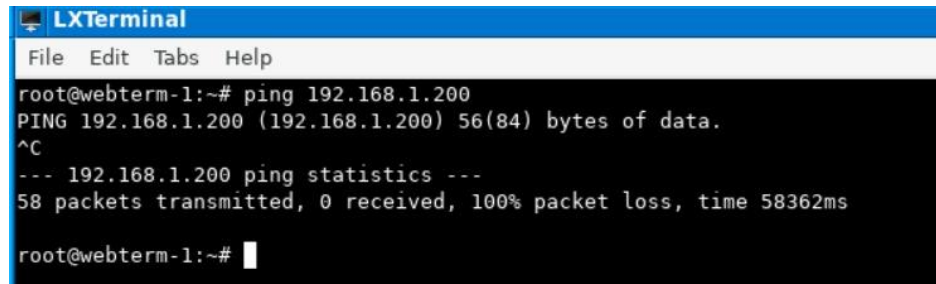
4. Referring to the following output, associate each field with the corresponding category in the table:
cookie=0x0, duration=124.897s, table=0, n_packets=14, n_bytes=728, priority=0 actions=CONTROLLER:65535

- Flow Table number: table=0
- Criteria priority: priority=0
- Selection criteria: No tiene
- Action to be taken: actions=CONTROLLER:65535
- Number of packages matching this criterion: n_packets=14

5. Add a flow rule on switch OVS1 to block ping requests between PC1 and Toolbox-web.
 - a. Execute the following command on OVS1. `sudo ovs-ofctl add-flow br0 "priority=10, icmp, nw_src=192.168.1.100, nw_dst=192.168.1.200, actions=drop"`
 - b. On the Local Machine, use the web interface to check that the rule has been added to OVS1.

```
OpenvSwitch-1:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=191.451s, table=0, n_packets=0, n_bytes=0,
priority=10,icmp,nw_src=192.168.1.100,nw_dst=192.168.1.200
actions=drop
```

- c. On PC1, execute a ping command to verify that the communication between the two devices is blocked

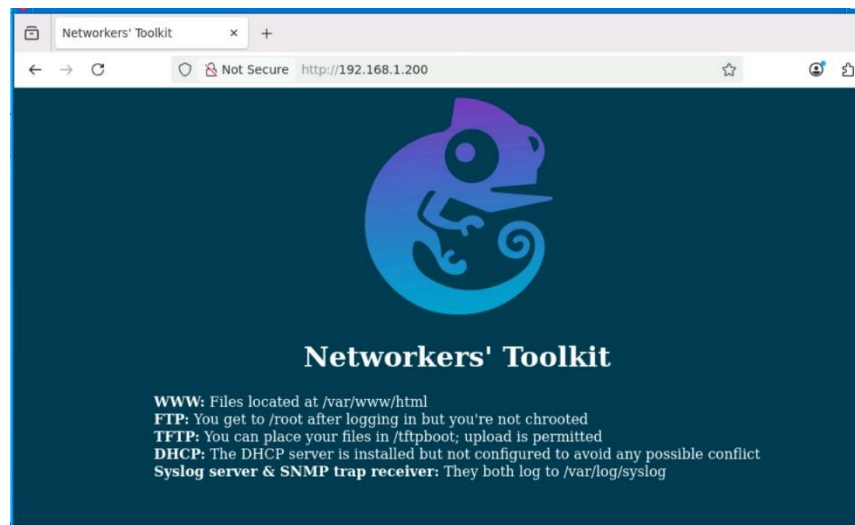


```
LXTerminal
File Edit Tabs Help
root@webterm-1:~# ping 192.168.1.200
PING 192.168.1.200 (192.168.1.200) 56(84) bytes of data.
^C
--- 192.168.1.200 ping statistics ---
58 packets transmitted, 0 received, 100% packet loss, time 58362ms

root@webterm-1:~#
```

```
OpenvSwitch-1:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=972.350s, table=0, n_packets=58, n_bytes=5684, priority=10,icmp,nw_src=192.168.1.100,nw_dst=192.168.1.200 actions=drop
```

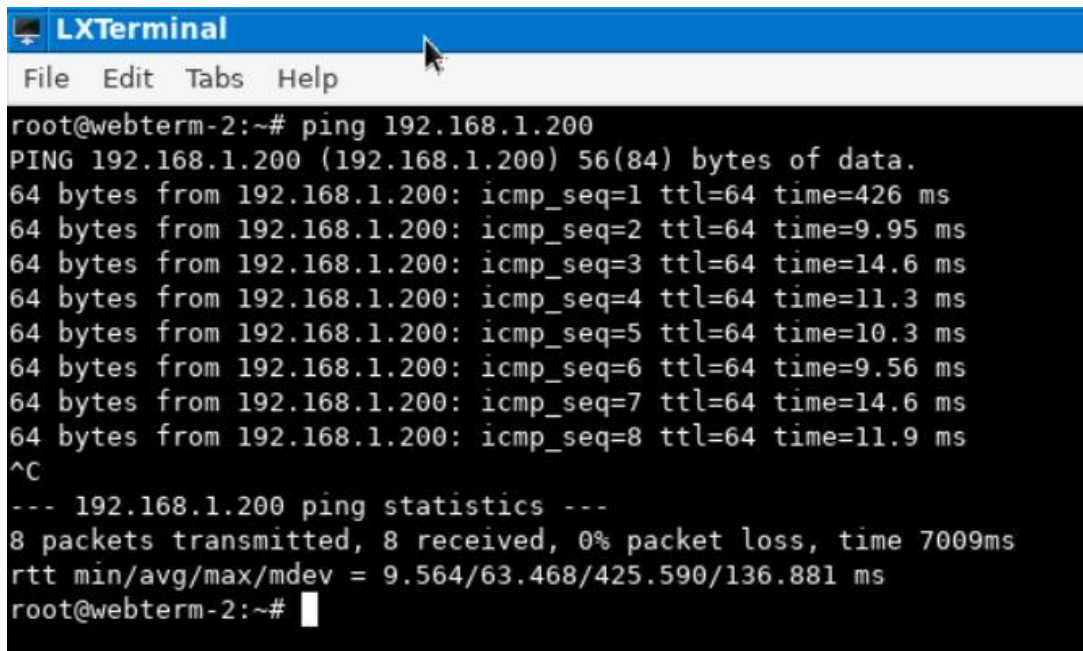
- d. On PC1, check that you have access to the Toolbox-web web page by entering the following URL: <http://192.168.1.200>



6. Add a flow rule on switch OVS2 to block TCP requests to 80 from PC2 to Toolbox-web.
 - a. Execute the following command on OVS2. `ovs-ofctl add-flow br0 "priority=10, tcp,nw_dst=192.168.1.200, tp_dst=80, actions=drop"`
 - b. On the Local Machine, use the web interface to check that the rule has been added to OVS2.

```
OpenvSwitch-2:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=65.959s, table=0, n_packets=0, n_bytes=0,
priority=10,tcp,nw_dst=192.168.1.200,tp_dst=80 actions=drop
```

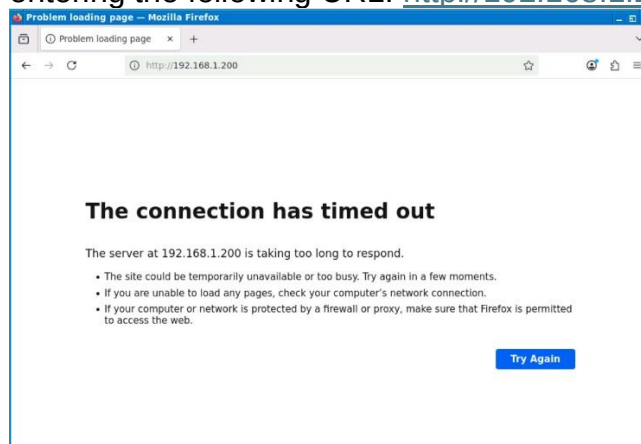
- c. On PC2, use a ping command to check that communication between the two elements is authorized.



```
LXTerminal
File Edit Tabs Help

root@webterm-2:~# ping 192.168.1.200
PING 192.168.1.200 (192.168.1.200) 56(84) bytes of data.
64 bytes from 192.168.1.200: icmp_seq=1 ttl=64 time=426 ms
64 bytes from 192.168.1.200: icmp_seq=2 ttl=64 time=9.95 ms
64 bytes from 192.168.1.200: icmp_seq=3 ttl=64 time=14.6 ms
64 bytes from 192.168.1.200: icmp_seq=4 ttl=64 time=11.3 ms
64 bytes from 192.168.1.200: icmp_seq=5 ttl=64 time=10.3 ms
64 bytes from 192.168.1.200: icmp_seq=6 ttl=64 time=9.56 ms
64 bytes from 192.168.1.200: icmp_seq=7 ttl=64 time=14.6 ms
64 bytes from 192.168.1.200: icmp_seq=8 ttl=64 time=11.9 ms
^C
--- 192.168.1.200 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7009ms
rtt min/avg/max/mdev = 9.564/63.468/425.590/136.881 ms
root@webterm-2:~#
```

- a. On PC2, verify that you can't access the Toolbox-web web page by entering the following URL: <http://192.168.1.200>



7. Add a flow rule on flow table number 1 of switch OVS3 to block TCP requests on port 80 from PC3 to Toolbox-web.
 - a. Execute the following command on OVS3. `ovs ofctl add flow br0 " table=1, priority=10, tcp,nw_dst=192.168.1.200, tp_dst=80, actions=drop"`
 - b. On the Local Machine, use the web interface to check that the rule has been added to OVS3.

```
OpenvSwitch-3:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=65.235s, table=1, n_packets=0, n_bytes=0,
priority=10,tcp,nw_dst=192.168.1.200,tp_dst=80 actions=drop
```

8. What is the purpose of using multiple flow tables in the OpenFlow protocol?
Dividir las reglas para cada tipo de paquete de una manera más ordenada y rápida en varias tablas a comparación de hacerlo en una sola.
9. Create a "Fast Failover" OpenFlow group table on the OVS4 switch to enable redundancy between ports 2 and 3. This will ensure that if port 2 fails, the switch will automatically switch to port 3.
 - a. Execute the following command on OVS4 `ovs-ofctl add-group br0 group_id=1,type=ff,bucket=watch_port:2, action=output:2, bucket=watch_port:3,action=output:3`
 - b. On the Local Machine, use the web interface to check that the rule has been added to OVS4.

```
OpenvSwitch-4:/# ovs-ofctl -O OpenFlow13 dump-groups br0
OFPST_GROUP_DESC reply (OF1.3) (xid=0x2):
```

```
group_id=1,type=ff,bucket=watch_port:eth1,actions=output:eth1,buc
ket=watch_port:eth2,actions=output:eth2
OpenvSwitch-4:/#
```

10. Create an OpenFlow group table of type "Select" on switch OVS2 to enable load balancing between ports 5 and 6. Execute the following command on OVS2.


```
ovs-ofctl add-group br0
group_id=1,type=select,bucket=output:5,bucket=output:6
```

SALIDA:

```
OpenvSwitch-2:/# ovs-ofctl -O OpenFlow13 add-group br0
group_id=1,type=select,bucket=output:5,bucket=output:6
OpenvSwitch-2:/# ovs-ofctl -O OpenFlow13 dump-groups br0
OFPST_GROUP_DESC reply (OF1.3) (xid=0x2):
group_id=1,type=select,bucket=actions=output:eth4,bucket=actions=out
put:eth5
```

11. Route all traffic destined to the 192.168.2.0/28 network through ports 5 and 6, so that it is processed by the group table already created.
 - a. Execute the following command on OVS2. `ovs-ofctl add-flow br0 "priority=100,ip,nw_dst=192.168.2.0/28,actions=group:1"`
 - b. On the Local Machine, use the web interface to verify that the two rules have been added to OVS2.

```
OpenvSwitch-2:/# ovs-ofctl -O OpenFlow13 add-flow br0
"priority=100,ip,nw_dst=192.168.2.0/28,actions=group:1"
OpenvSwitch-2:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=21.118s, table=0, n_packets=0, n_bytes=0,
priority=100,ip,nw_dst=192.168.2.0/28 actions=group:1
```

12. Configure QoS on port 4 of OVS3 to limit throughput to 100 Mbps for the 172.16.1.0/24 network
 - a. Execute the following command on OVS3 to create a q0 queue and limit the bandwidth: `ovs-vsctl set port eth4 qos=@newqos -- --id=@newqos create qos type=linux-htb other-config:max-rate=100000000 queues=0=@q0 -- --id=@q0 create queue other-config:min-rate=8000000 other-config:max-rate=100000000`

SALIDA:

```
OpenvSwitch-3:/# ovs-vsctl set port eth4 qos=@newqos -- --
id=@newqos create qos type=linux-htb other-config:max-
rate=100000000 queues=0=@q0 -- --id=@q0 create queue other-
config:min-rate=8000000 other-config:max-rate=100000000
72c0605c-38ec-464c-8cce-e8dd285cafb2
20d451b1-3c7f-431d-8736-0d8027c6b138
```

```
OpenvSwitch-3:/# ovs-vsctl list qos
 _uuid          : 72c0605c-38ec-464c-8cce-e8dd285cafb2
external_ids    : {}
other_config    : {max-rate="100000000"}
queues         : {0=20d451b1-3c7f-431d-8736-0d8027c6b138}
type           : linux-htb
OpenvSwitch-3:/# ovs-vsctl list queue
 _uuid          : 20d451b1-3c7f-431d-8736-0d8027c6b138
dscp           : []
external_ids    : {}
other_config    : {max-rate="100000000", min-rate="8000000"}
OpenvSwitch-3:/#
```

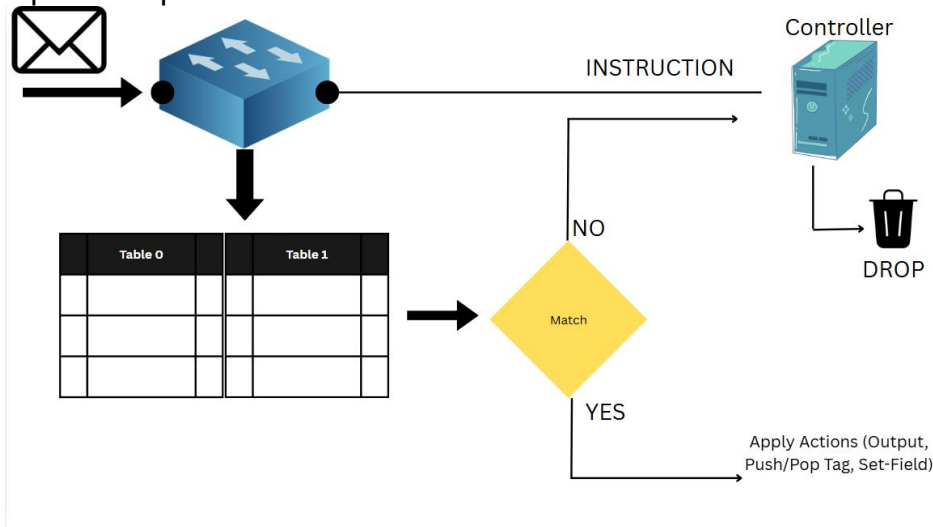
- b. Execute the following command on OVS3 to apply the q0 queue to the traffic of the 172.16.1.0/24 network: `ovs-ofctl add-flow br0 "table=2, priority=100, ip, nw_dst=172.16.1.0/24, actions=set_queue:0,normal"`
- c. On the Local Machine, use the web interface to verify that the two rules have been added to OVS2.

```

OpenvSwitch-3:/# ovs-ofctl -O OpenFlow13 add-flow br0 "table=2,
priority=100, ip, nw_dst=172.16.1.0/24, actions=set_queue:0,normal"
OpenvSwitch-3:/# ovs-ofctl -O OpenFlow13 dump-flows br0
cookie=0x0, duration=44.591s, table=2, n_packets=0, n_bytes=0,
priority=100,ip,nw_dst=172.16.1.0/24 actions=set_queue:0,NORMAL

```

13. In the context of OpenFlow and SDN networks, what is the description of a "table-miss" rule in an OpenVSwitch switch?
Respuesta inciso C: A "table-miss" rule is a special rule that defines the action to be taken for packets that don't match any other rule in the flow table.
14. What is the "table-miss" value of OVS switches?
La prioridad 0
15. Referring to the roles of the flow and group tables you have studied, please create a flowchart illustrating the processing of a packet by the OpenFlow protocol.



PARTE D

Referring to the following log messages, indicate the corresponding OpenFlow message type for each line:

OpenFlow message type	Description
CTR => OVS	A message sent by the controller
OVS => CTR	A message sent by the OVS
CTR <=> OVS	A symmetrical message

Log messages	Message type
2023-09-28 14:15:03.257 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Received OpenFlow HELLO message from OVS (192.168.1.20) to Controller (192.168.1.10). Version: OF_13	CTR <=> OVS
2023-09-28 14:16:05.123 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Received OpenFlow SET CONFIG message from OVS (192.168.1.20) to Controller (192.168.1.10). Configuration set to drop fragments.	OVS => CTR
2023-09-28 14:16:18.556 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Sent OpenFlow PACKET-OUT message from Controller (192.168.1.10) to OVS (192.168.1.20). Egress Port: 2, Data Length: 64 bytes	CTR => OVS
2023-09-28 14:16:12.890 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Received OpenFlow PACKET-IN message from OVS (192.168.1.20) to Controller (192.168.1.10). Ingress Port: 1, Reason: NO_MATCH.	OVS => CTR
2023-09-28 14:16:15.303 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Sent OpenFlow FLOW-MOD message from Controller (192.168.1.10) to OVS (192.168.1.20). Match: Ingress Port 1, Action: Output Port 2.	CTR => OVS

2023-09-28 14:16:21.897 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Sent OpenFlow FLOW-MOD message from Controller (192.168.1.10) to OVS (192.168.1.20). Match: Source IP 192.168.1.100, Action: Drop.	CTR => OVS
2023-09-28 14:16:05.123 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Received OpenFlow SET CONFIG message from openflow:1 (br0) to Controller (192.168.1.10). Configuration set to drop fragments.	OVS => CTR
2023-09-28 14:16:12.890 [DEBUG] [org.ryu.openflowjava.protocol.impl] - Received OpenFlow PACKET-IN message from openflow:1 (br0) to Controller (192.168.1.10). Ingress Port: 1, Reason: NO_MATCH	OVS => CTR